

GARAGE MANAGEMENT SYSTEM

A PROJECT REPORT

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BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



**UNIVERSITY COLLEGE OF ENGINEERING,
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NOV/DEC 2025

ABSTRACT

The **Garage Management System** is a Salesforce-based cloud application developed to automate and streamline garage operations such as **vehicle service scheduling, billing, payment tracking, and customer feedback management**.

The system manages the entire workflow — from customer appointment booking to service completion and payment processing — with the help of **Apex classes, triggers, flows, dashboards, and reports**.

The main goal of this project is to reduce manual effort, improve efficiency, and provide transparency between customers and garage administrators.

Key tools used include:

- Salesforce CRM Platform
- Flows and Automation
- Apex Trigger and Handler
- Reports and Dashboards

This system provides a complete digital transformation for garage businesses, ensuring faster service, accurate billing, and enhanced customer satisfaction.

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CHAPTER 1 – INTRODUCTION

1.1 Overview

- The Garage Management System is designed to automate daily operations of a vehicle service center. It enables garage owners to manage appointments, track vehicle services, update payment status, and collect feedback efficiently.
- Using Salesforce CRM, the system maintains various records — **Customer Details, Appointments, Service Records, and Billing & Feedback** — all interconnected to provide a seamless workflow.

1.2 Objectives

- To create an automated system to manage garage operations.
- To implement Salesforce features such as Objects, Flows, Apex Triggers, and Dashboards.
- To enable real-time tracking of customer services and billing.
- To enhance user satisfaction through feedback and report analytics.

1.3 Scope

The project covers:

- Appointment scheduling and tracking
- Automatic billing calculation and payment update
- Feedback management
- Report and dashboard generation for performance analysis

1.4 Existing vs Proposed System

Existing System	Proposed System (Salesforce)
Manual record keeping	Fully automated digital system
High chance of human errors	Data integrity maintained
Time-consuming report generation	Real-time dashboards and analytics
No customer feedback tracking	Integrated feedback and quality monitoring

CHAPTER 2 – PROJECT IDEATION (Phase 1)

2.1 Problem Identification

Traditional garages face challenges such as missing service records, manual calculations, and lack of transparency. Customers often have difficulty tracking service progress or payment details.

2.2 Idea Description

The idea focuses on building a **centralized Salesforce-based Garage Management System** to manage all processes — from service booking to billing and customer satisfaction — through automation.

2.3 Feasibility

Salesforce provides ready-made CRM tools, automation, and reporting capabilities. It requires no additional infrastructure, making it cost-effective and scalable.

2.4 Innovation

- Use of **record-triggered flows** for payment updates
- **Automated email alerts** to customers after service completion
- **Dashboard visualization** of service ratings and payments

CHAPTER 3 – REQUIREMENT ANALYSIS (Phase 2)

3.1 Functional Requirements

- Add, update, and delete customer records.
- Automatically calculate service amount based on selected services.
- Update payment and service status dynamically.
- Generate analytical reports and dashboards.

3.2 Non-Functional Requirements

- Data security ensured by Salesforce cloud.
- User-friendly interface.
- Scalability for future improvements.

3.3 User Roles

- Admin: Manages all operations.
- Service Staff: Updates service records.
- Customer: Views payment and feedback updates.

3.4 Data Flow & ER Diagram

- Objects: Customer → Appointment → Service Records → Billing & Feedback
- Relationship: Master-detail and Lookup relationships connect all entities.

CHAPTER 4 – PROJECT DESIGN (Phase 3)

4.1 Schema Builder

Visualizes relationships between objects like Customer, Appointment, Service Records, and Billing.

4.2 Process Flow

1. Customer books appointment.
2. Service record created.
3. Billing and payment generated automatically.
4. Email alert sent to customer.

4.3 UI/UX Design

Simple, user-friendly Salesforce layout using custom objects and tabs for easy navigation.

CHAPTER 5 – PROJECT DEVELOPMENT (Phase 4)

5.1 Implementation Details

- **Objects Created:** Customer Details, Appointment, Service Records, Billing & Feedback
- **Fields:** Custom fields like Payment Status, Service Amount, and Rating for Service

5.2 Apex Code (Handler)

```
public class AmountDistributionHandler {

    public static void amountDist(list<Appointment__c> listApp){

        for(Appointment__c app : listApp){

            if(app.Maintenance_service__c && app.Repairs__c && app.Replacement_Parts__c){

                app.Service_Amount__c = 10000;

            } else if(app.Maintenance_service__c && app.Repairs__c){

                app.Service_Amount__c = 5000;

            } else if(app.Repairs__c){

                app.Service_Amount__c = 3000;

            } else if(app.Replacement_Parts__c){

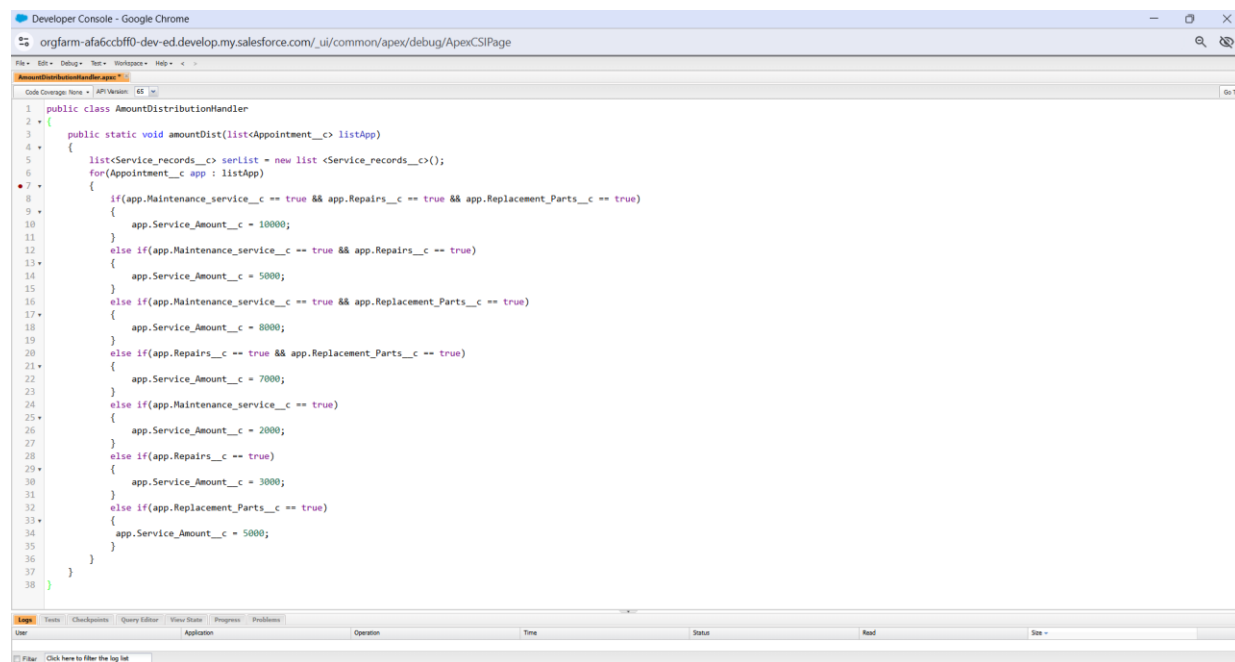
                app.Service_Amount__c = 5000;

            }

        }

    }

}
```



5.3 Trigger

```
trigger AmountDistribution on Appointment__c (before insert, before update) {  
    if(trigger.isBefore && (trigger.isInsert || trigger.isUpdate)){  
        AmountDistributionHandler.amountDist(trigger.new);  
    }  
}
```

5.4 Flows

- **Flow 1:** Updates payment details when status is “Completed”.
- **Flow 2:** Updates service status when Quality Check = True.

5.5 Reports & Dashboards

- **Report:** “Service Information Report” with grouped data by Payment Status and Rating.
- **Dashboard:** “Customer Review Dashboard” with weekly summary chart.

CHAPTER 6 – PROJECT DOCUMENTATION (Phase 5)

6.1 Execution Steps

- Created objects, relationships, and fields.
- Implemented triggers and flows.
- Tested validation rules.
- Created dashboards and reports.

6.2 GitHub & SmartInternz Links

- **GitHub Repository:** <https://github.com/sirajnish0604-prog/Garage-Management-System>
- **SmartInternz Workspace:** https://nme.smartinternz.com/Student/guided_project_workspace/3739

6.3 Challenges & Solutions

Challenge	Solution
“Done” button not visible in Flow screen	Adjusted screen zoom & saved directly
Picklist value update error	Corrected field mapping
Missing object fields	Verified API names

CHAPTER 7 – PROJECT DEMONSTRATION & RESULTS

7.1 Demo Flow

When billing is completed, flow updates Payment Status and sends email alert to customer.

7.2 Outputs

- Billing and feedback automatically updated.
- Email alert sent successfully.
- Dashboard displays service rating summary.

CHAPTER 8 – CONCLUSION & FUTURE ENHANCEMENT

8.1 Conclusion

The Garage Management System successfully automates garage services using Salesforce automation, triggers, and flows, ensuring efficient service and data tracking.

8.2 Limitations

- Limited to Salesforce environment.
- Requires internet connectivity.

8.3 Future Enhancements

- Add customer login portal.
- Integrate mobile app version.
- Include AI-based service prediction.

APPENDIX – SCREENSHOTS

Garage Management App home

etud Home Object Manager

App Launcher [Visit AppExchange](#)

▼ All Apps


[Garage Management Application](#)

▼ All Items

[Capability Navigator](#)
[Engagement Channel Types](#)
[Payment Gateway Logs](#)

[Data Use Legal Basis](#)
[Legal Entities](#)
[Payment Gateways](#)

Cloud Search...

Garage Management Appl... Customer Details Appointments Service records Billing details and feedback Reports Dashboards Quality Checks

Customer Details

Recently Viewed

12 items • Updated 10 minutes ago

[New](#) [Import](#) [Change Owner](#) [Assign Label](#)

	☐ Customer Name	
1	☐ Mac	▼
2	☐ Priya	▼
3	☐ rathi	▼
4	☐ karthi	▼
5	☐ Natthar	▼
6	☐ siva	▼
7	☐ roja	▼
8	☐ mani	▼
9	☐ susmitha	▼
10	☐ sales person	▼
11	☐ sheak	▼
12	☐ durga	▼

Appointments

Recently Viewed ▾ ⓘ

15 items • Updated a few seconds ago

Search this list...

☐	Appointment Name	
1	app-016	⌵
2	app-015	⌵
3	app-014	⌵
4	app-013	⌵
5	app-012	⌵
6	app-010	⌵
7	app-009	⌵
8	app-008	⌵
9	app-007	⌵
10	app-006	⌵
11	app-005	⌵
12	app-004	⌵
13	app-003	⌵
14	app-002	⌵
15	app-001	⌵

Service records

Recently Viewed ▾ ⓘ

12 items • Updated a few seconds ago

Search this list...

☐	Service records Name	
1	ser-012	⌵
2	ser-011	⌵
3	ser-004	⌵
4	ser-010	⌵
5	ser-009	⌵
6	ser-008	⌵
7	ser-007	⌵
8	ser-006	⌵
9	ser-005	⌵
10	ser-003	⌵
11	ser-002	⌵
12	ser-001	⌵

Billing details and feedback

Recently Viewed ▾ ⓘ

10 items • Updated a few seconds ago

Search this list...

☐	Billing details and feedback Name	
1	bill-016	⌵
2	bill-015	⌵
3	bill-013	⌵
4	bill-012	⌵
5	bill-011	⌵
6	bill-010	⌵
7	bill-007	⌵
8	bill-005	⌵
9	bill-004	⌵
10	bill-001	⌵

Garage Management Appl...

Customer Details Appointments Service records Billing details and feedback **Reports** Dashboards Quality Checks

Reports

Recent

1 item

REPORTS

Report Name	Description	Folder	Created By	Created On	Subscribed
New Service Information Report	This report shows customer service details, payment, and ratings.	Garage Management Folder	SIRAJ NISHA S	11/1/2025, 1:00 AM	

Created by Me

Private Reports

Public Reports

All Reports

FOLDERS

All Folders

Created by Me

Shared with Me

FAVORITES

All Favorites

Fields & Relationships for an object

Setup Home **Object Manager**

SETUP > OBJECT MANAGER

Appointment

Details

Fields & Relationships

11 Items, Sorted by Field Label

Q, Quick Find

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Appointment Date	Appointment_Date__c	Date		
Appointment Name	Name	Auto Number		✓
Created By	CreatedById	Lookup(User)		
Customer Details	Customer_Details__c	Lookup(Customer Details)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Maintenance service	Maintenance_service__c	Checkbox		
Owner	OwnerId	Lookup(User,Group)		✓
Repairs	Repairs__c	Checkbox		
Replacement Parts	Replacement_Parts__c	Checkbox		

Setup Home **Object Manager**

SETUP > OBJECT MANAGER

Billing details and feedback

Details

Fields & Relationships

8 Items, Sorted by Field Label

Q, Quick Find

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Billing details and feedback Name	Name	Auto Number		✓
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Payment Paid	Payment_Paid__c	Currency(18, 0)		
Payment Status	Payment_Status__c	Picklist		
Rating for service	Rating_for_service__c	Text(1)		
Service records	Service_records__c	Lookup(Service records)		✓

Setup Home **Object Manager**

Customer Details

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules

Fields & Relationships
 6 Items, Sorted by Field Label

Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer Name	Name	Text(80)		✓
Gmail	Gmail__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone no	Phone_no__c	Phone		

Service records

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Fields & Relationships
 9 Items, Sorted by Field Label

Q, Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Appointment	Appointment__c	Lookup(Appointment)		✓
Appointment Date	Appointment_Date__c	Date		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Quality Check Status	Quality_Check_Status__c	Checkbox		
service date	service_date__c	Formula (Date)		
Service records Name	Name	Auto Number		✓

Setup Home **Object Manager**

Appointment

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules

Appointment Custom Field

Vehicle number plate

Back to Appointment

Validation Rules (1)

Help for this Page

Custom Field Definition Detail

Edit Set Field Level Security View Field Accessibility Where is this used?

Field Information

Field Label	Vehicle number plate	Object Name	Appointment
Field Name	Vehicle_number_plate	Data Type	Text
API Name	Vehicle_number_plate__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	SIRAJ NISHA S. 10/31/2025, 6:06 AM	Modified By	SIRAJ NISHA S. 10/31/2025, 7:03 AM

General Options

Required	✓
Unique	✓
Case Sensitive	
External ID	✓

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Appointment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Appointment Custom Field

Service Amount

Back to Appointment

Validation Rules (0)

Help for this Page

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	Service Amount	Object Name	Appointment
Field Name	Service_Amount	Data Type	Currency
API Name	Service_Amount__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	SIRAJ NISHA S. 10/31/2025, 6:03 AM	Modified By	SIRAJ NISHA S. 10/31/2025, 6:03 AM

General Options

Required	☐
Default Value	

Currency Options

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Appointment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Appointment Custom Field

Appointment Date

Back to Appointment

Validation Rules (0)

Help for this Page

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	Appointment Date	Object Name	Appointment
Field Name	Appointment_Date	Data Type	Date
API Name	Appointment_Date__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	SIRAJ NISHA S. 10/31/2025, 1:37 AM	Modified By	SIRAJ NISHA S. 10/31/2025, 2:02 AM

General Options

Required	☒
Default Value	

Validation Rules

New

Validation Rules Help

Validation Rule

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Appointment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Appointment Validation Rule

Back to Appointment

Help for this Page

Validation Rule Detail

Edit

Clone

Rule Name	Vehicle	Active	☒
Error Condition Formula	NOT(REGEX(Vehicle_number_plate__c , "[A-Z][0-9][2][A-Z][0-9][4]"))		
Error Message	Please enter valid number vehicle	Error Location	Vehicle number plate
Description			
Created By	SIRAJ NISHA S. 10/31/2025, 7:07 AM	Modified By	SIRAJ NISHA S. 10/31/2025, 7:07 AM

Edit

Clone

SETUP > OBJECT MANAGER

Billing details and feedback

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Billing details and feedback Validation Rule

[Back to Billing details and feedback](#)

[Help for this Page](#)

Validation Rule Detail

[Edit](#) [Clone](#)

Rule Name rating_should_be_less_than_5

Active

✓

Error Condition Formula NOT(REGEX(Rating_for_service__c, '1-5[1]*'))

Error Location

Rating for service

Error Message rating should be from 1 to 5

Description

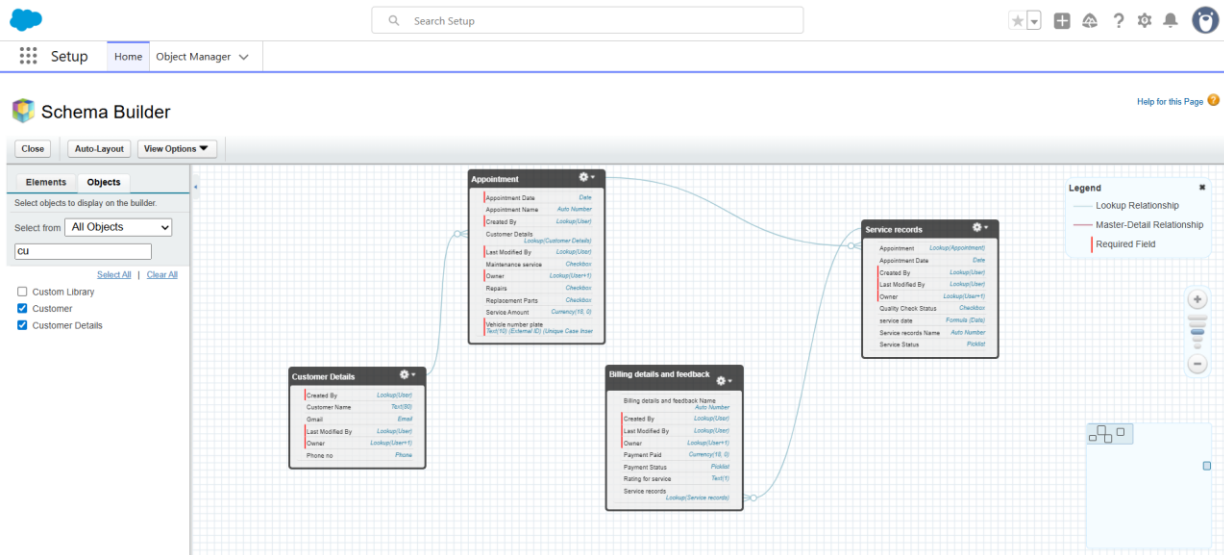
Created By SIRAJ NISHA S, 10/31/2025, 7:13 AM

Modified By

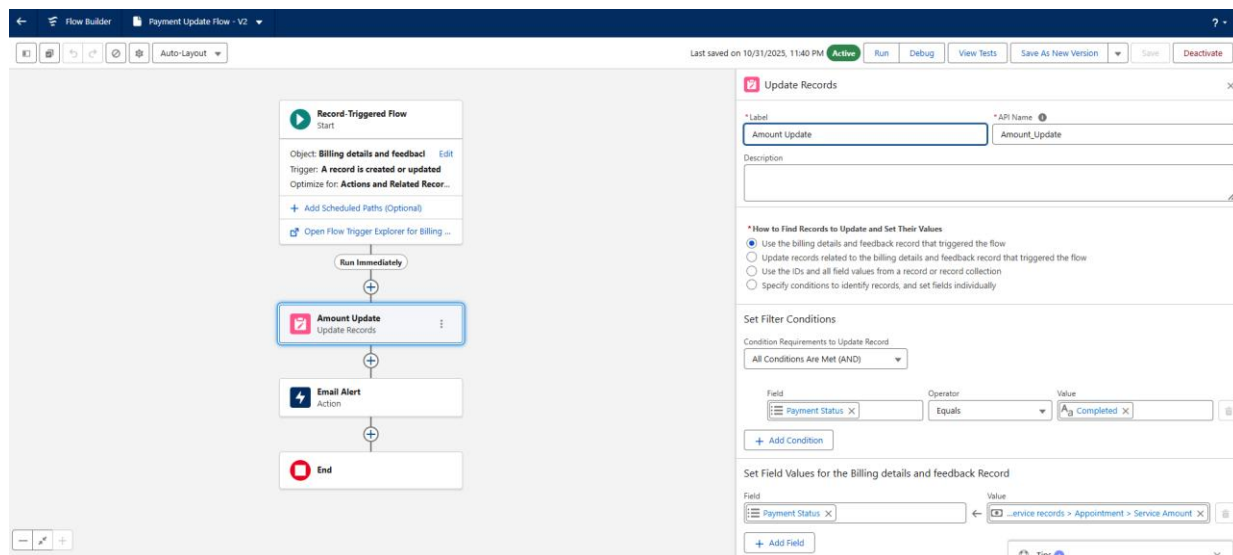
SIRAJ NISHA S, 10/31/2025, 7:13 AM

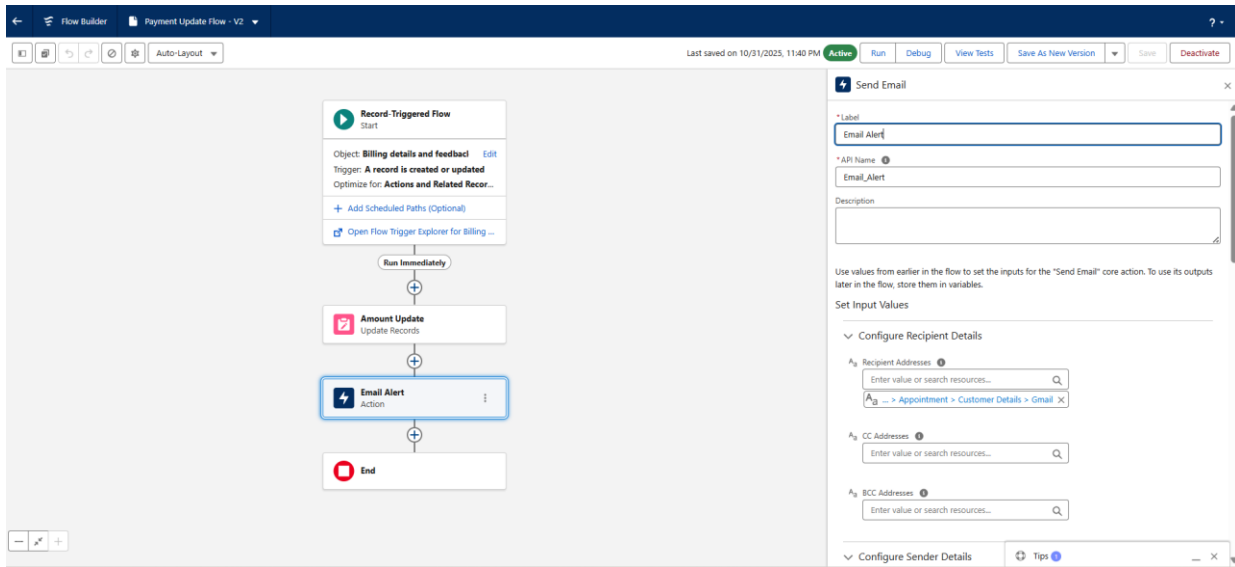
[Edit](#) [Clone](#)

Schema Builder

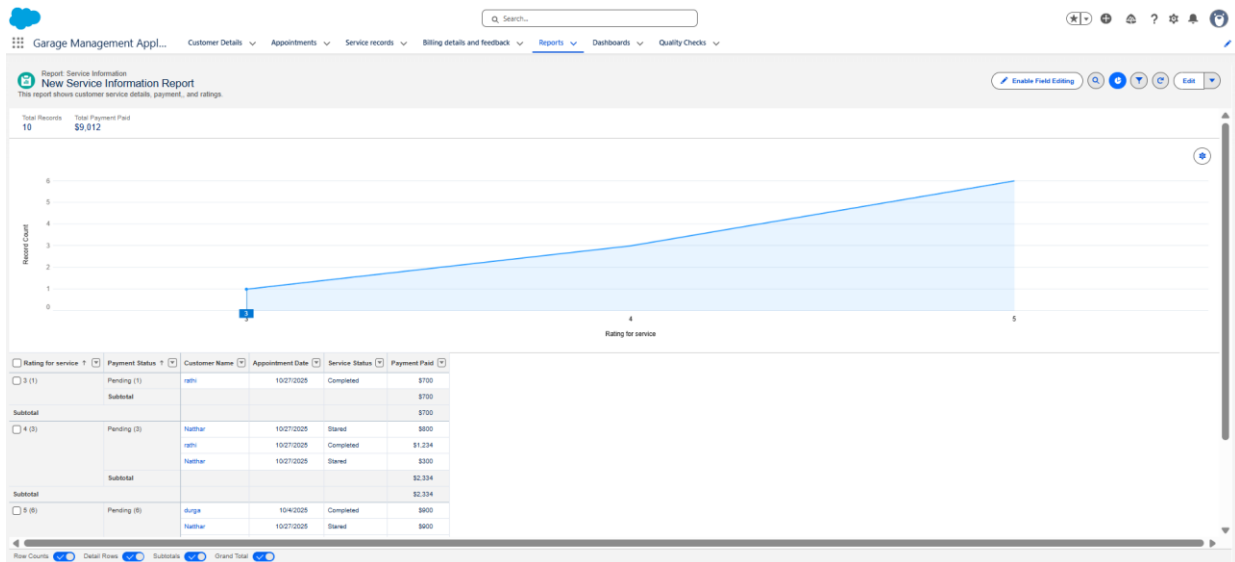


Automation Flow (Flow Builder)





Report preview



Dashboard

Garage Management Appl...

Search...

Customer Details Appointments Service records Billing details and feedback Reports Dashboards Quality Checks

Dashboards

Recent

1 item

Search recent dashboards... New Dashboard New Folder

DASHBOARDS	Dashboard Name	Description	Folder	Created By	Created On	Subscribed
Recent	Customer review	Garage Management Review Report	Service Rating Dashboard	SIRAJ NISHA S	11/1/2025, 1:28 AM	✓

Created by Me

Private Dashboards

All Dashboards

FOLDERS

All Folders

Created by Me

Shared with Me

FAVORITES

All Favorites



Garage Management Appl... Customer Details Appointments Service records Billing details and feedback Reports Dashboards



Dashboard

Customer review

Garage Management Review Report

⚠ Last refreshed 1 day ago. Refresh this dashboard to see the latest data.

As of Nov 3, 2025, 2:00 PM Viewing as SIRAJ NISHA S

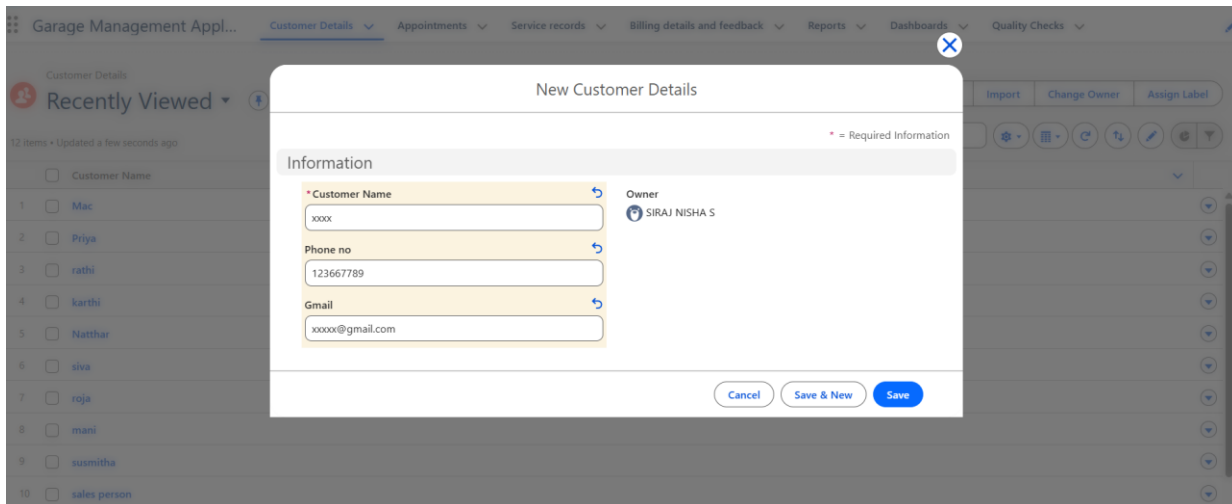
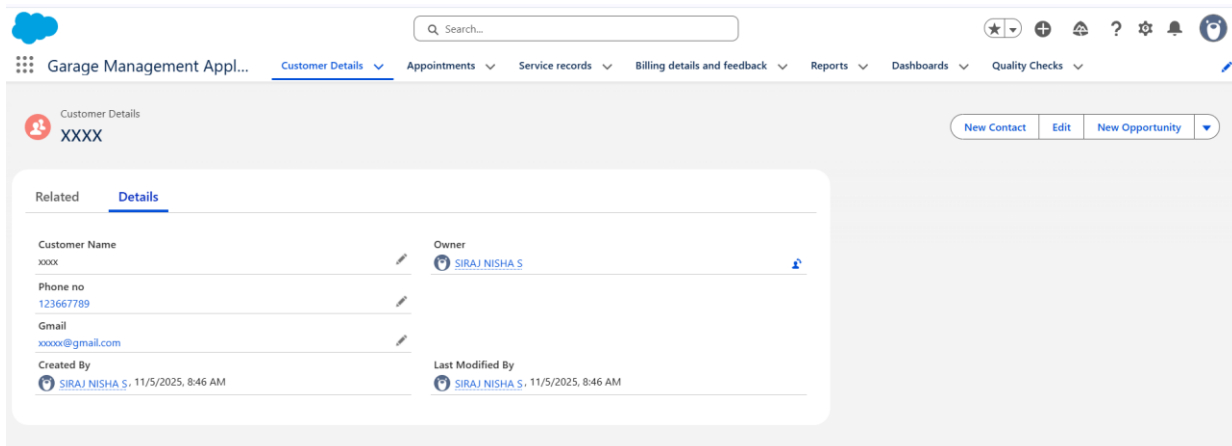
New Service Information Report



[View Report \(New Service Information Report\)](#)

As of Nov 3, 2025, 2:00 PM

Record Creation (show working flow)

Garage Management Appl... Customer Details Appointments

Search...

New Appointment

Information

Appointment Name

Customer Details

xxxx

Appointment Date

11/10/2025

Maintenance service

Repairs

Replacement Parts

Service Amount

\$800

Vehicle number plate

TN10AB0267

Owner

SIRAJ NISHA S

Cancel Save & New Save

Garage Management Appl... Customer Details Appointments Service records Billing details and feedback Reports Dashboards Quality Checks

Search...

Appointment app-018

New Contact Edit New Opportunity

Related Details

Appointment Name

app-018

Owner

SIRAJ NISHA S

Customer Details

xxxx

Appointment Date

11/10/2025

Maintenance service

Repairs

Replacement Parts

Service Amount

\$10,000

Vehicle number plate

TN10AB0267

Created By

SIRAJ NISHA S, 11/5/2025, 8:48 AM

Last Modified By

SIRAJ NISHA S, 11/5/2025, 8:48 AM

Garage Management Appl... Customer Details Appointments Service records Billing details and feedback Reports Dashboards Quality Checks

Search...

New Service records

Information

Service records Name

Appointment Date

11/10/2025

Appointment

app-015

Quality Check Status

Service Status

Stared

Owner

SIRAJ NISHA S

Cancel Save & New Save

Garage Management Appl...

Customer Details

Appointments

Service records

Billing details and feedback

Reports

Dashboards

Quality Checks

Service records

ser-013

New Contact

Edit

New Opportunity

Related

Details

Service records Name

ser-013

Appointment Date

11/10/2025

Appointment

ser-013

Quality Check Status

Service Status

Completed

Service date

11/5/2025

Created By

SIRAJ NISHA S

Owner

SIRAJ NISHA S

Last Modified By

SIRAJ NISHA S

Garage Management Appl...

Customer Details

Appointments

Service records

Billing details and feedback

Reports

Dashboards

Quality Checks

Billing details and feedback

Recently Viewed

New

Import

Change Owner

Assign Label

11 items • Updated a few seconds ago

Billing details and feedback Name

1 bill-017

2 bill-016

3 bill-015

4 bill-013

5 bill-012

6 bill-011

7 bill-010

8 bill-007

9 bill-003

10 bill-004

11 bill-001

New Billing details and feedback

Information

Billing details and feedback Name

Service records

ser-013

Payment Paid

Rating for service

5

Payment Status

Pending

Owner

SIRAJ NISHA S

Cancel

Save & New

Save

Garage Management Appl...

Customer Details

Appointments

Service records

Billing details and feedback

Reports

Dashboards

Quality Checks

Billing details and feedback

bill-018

Billing details and feedback "bill-018" was created.

New Contact

Edit

New Opportunity

Related

Details

Billing details and feedback Name

bill-018

Service records

ser-013

Payment Paid

Rating for service

5

Payment Status

Pending

Created By

SIRAJ NISHA S

Owner

SIRAJ NISHA S

Last Modified By

SIRAJ NISHA S

Garage Management Appl... Customer Details Appointments Service records Billing details and feedback Reports Dashboards Quality Checks

Quality Checks Recently Viewed

0 items • Updated a few seconds ago

New Quality Check

* = Required Information

Information

Auto Number

Service records

ser-013

Quality Check Status

Feedback

Owner

SIRAJ NISHA S

Cancel Save & New Save

Quality Check QC-001

Quality Check "QC-001" was created.

Related Details

Auto Number QC-001

Service records ser-013

Quality Check Status

Feedback

Owner

SIRAJ NISHA S

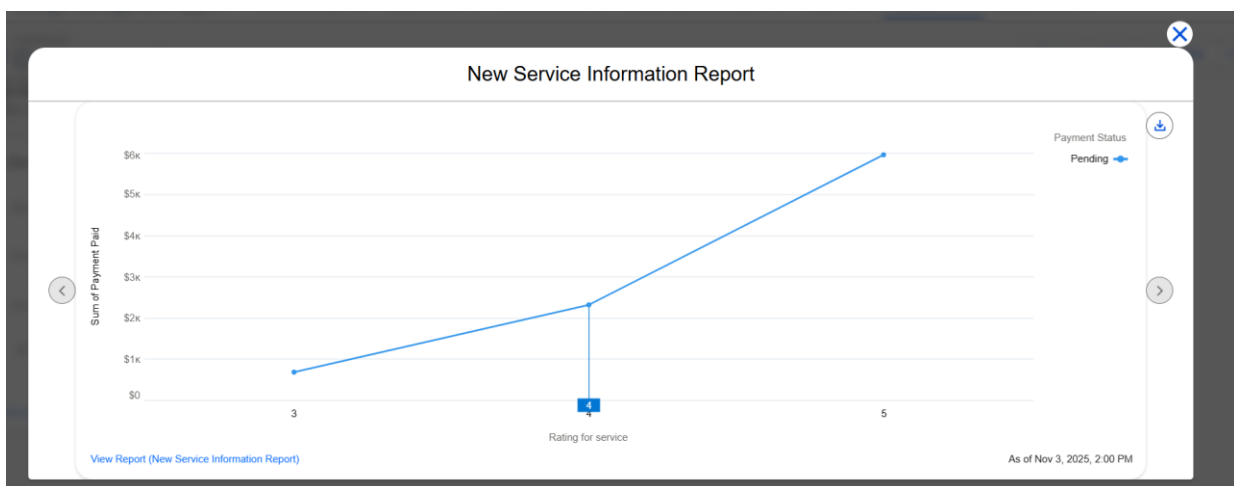
Created By

SIRAJ NISHA S, 11/5/2025, 8:54 AM

Last Modified By

SIRAJ NISHA S, 11/5/2025, 8:54 AM

Final UI & outputs



REFERENCES

1. Salesforce Trailhead Documentation — <https://trailhead.salesforce.com>
2. SmartInternz Naan Mudhalvan Portal — <https://nme.smartinternz.com>
3. Salesforce Developer Guide — <https://developer.salesforce.com>
4. GitHub Repository — <https://github.com/sirajnish0604-prog/Garage-Management-System>